Interview Prep Platform

AI-Powered Interview Practice with Intelligent Scoring & Feedback

SWE | Data Science | PM | Behavioral

Freeman Yiu · Youcheng Taing

Executive Summary

Business Concept

AI-powered web platform for interview practice with real-time scoring, feedback, and progress tracking per job role.

Target Market

CS students, bootcamp grads, and early-career engineers preparing for SWE, data science, and PM interviews.

Proposed Solution

Rubric-based + embedding similarity scoring with multi-dimensional feedback, no fine-tuning required.

Team Structure

2-person team: Backend & ML Lead (Youcheng) + Frontend & AI Systems Lead (Freeman). 9-week timeline.

Target Market

CS Students

University and bootcamp students preparing for internship and new grad interviews

Early-Career Engineers

0–3 years experience looking to level up or switch companies

Career Switchers

Non-CS professionals transitioning into software engineering or data science roles

Job Seekers

Active applicants who want structured, measurable practice with feedback

Universities

Career centers and CS departments offering guided interview prep resources

Solution Overview

Role Selection

User picks job role (SWE, Data Science, PM, Behavioral). System loads curated question bank for that role.

Rubric-Based Scoring

Each question has predefined rubric components. Answers are embedded and scored on coverage of key concepts.

Multi-Dimensional Feedback

4-axis scoring: Technical Depth · Clarity · Completeness · Structure. Score 0–100 per dimension.

Answer History

Every submission stored. Users track improvement over time with per-question history and score trends.

Improvement Suggestions

Missing rubric elements surfaced as actionable feedback. Clear explanation of what was strong and what was missing.

No Fine-Tuning Needed

Built on sentence-transformers + cosine similarity. Fast, free, runs locally. Impressive ML depth without complexity.

Team Structure

Team Member 1 — Youcheng Taing

Full-Stack & Backend Lead

- Backend development with FastAPI
- PostgreSQL database design & SQLAlchemy ORM
- API design, auth (JWT), and system integration
- ML pipeline: embedder, scorer, keyword extractor
- Docker + deployment setup

Team Member 2 — Freeman Yiu

Frontend & AI Systems Lead

- React UI development
- Score dashboard and radar chart visualization
- RAG pipeline and embedding integration
- Sentence-transformers AI model integration
- Deployment on Render with Docker

Technology Stack

Frontend

- React
- Recharts — radar + progress charts
- Tailwind CSS
- TypeScript

Backend

- FastAPI (Python)
- PostgreSQL + SQLAlchemy
- Alembic migrations
- JWT Authentication

AI / ML

- sentence-transformers (all-MiniLM-L6-v2)
- Cosine similarity scoring
- Keyword coverage extractor
- Multi-dim score engine

Infrastructure

- Docker + docker-compose
- Render deployment
- Redis (embedding cache — L3)
- GitHub Actions CI

9-Week Development Timeline

Wks 1–3	Wks 4–5	Wks 6–7	Wks 8–9
Backend + DB	**ML Pipeline**	**Frontend**	**Integration + Deploy**
FastAPI setup	Embedder	Role selection UI	Docker compose
DB schema	Cosine scorer	Q&A interface	Render deploy
Auth (JWT)	Keyword extractor	Score dashboard	QA + polish
Questions API	Multi-dim scoring	History page	Demo prep

Key Benefits

01 Explainable Scoring

Users know exactly why they scored low. Missing rubric elements surfaced as actionable steps — not a black box number.

02 No Fine-Tuning Overhead

sentence-transformers runs locally, free of charge. Impressive ML depth without GPU costs or training data collection.

03 Multi-Dimensional Insight

Technical Depth, Clarity, Completeness, and Structure scored independently. More useful than a single score.

04 Progress Tracking

Full answer history stored. Radar charts and trend lines show improvement over weeks of practice.

05 Role-Specific Practice

Curated question banks per role. SWE, Data Science, PM, and Behavioral tracks with relevant rubrics.

06 Production-Ready Architecture

FastAPI + Postgres + Docker. Auth, caching, and deployment built in. Scales from MVP to real product.

Ready to Build
Interview Intelligence

Let's discuss how we bring this to life in 9 weeks.

Questions & Discussion